

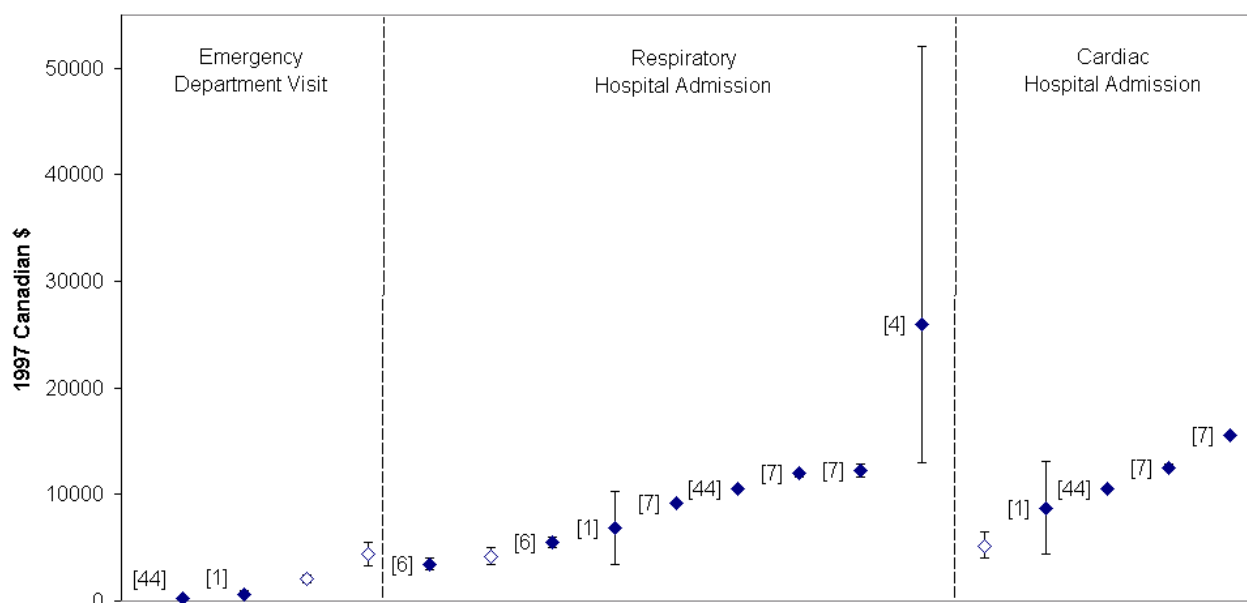


Figure 1

Comparison of valuation estimates for hospital admissions and emergency department visits with earlier studies. Legend: Labels denote reference number. Open symbols denote estimates from the present study. See Table 6 (additional file 1) for additional details on individual estimates.

Limitations of our approach

While we believe our results represent an advance over earlier estimates, there are several limitations which could be addressed in future work in this area. The model we employed for stated preference values was non-linear with respect to duration of the illness episode, implying diminishing marginal utility for reduced morbidity with respect to duration [28]. However, we assumed simple additivity of the individual blocks of time constituting each segment of activity restriction of varying severity in the context of an illness episode. This was necessitated by the finding in the context of the original stated preference survey, that presenting individuals with illness episode scenarios involving an evolution of level of activity restriction was overly burdensome for respondents. Our assumption of simple additivity is likely to lead to overestimates of total value relative to the observation in the original study of diminishing marginal utility with respect to duration [28]. The extent of bias requires further empirical evaluation. Additional development work would be needed to determine how best to communicate this information to re-

spondents in a straightforward yet accurate way, and to elicit valuation information appropriately.

Our analysis relied on data from two studies carried out in single centres (Saint John and Toronto). While we found that our results were not sensitive to demographic and health status differences between the original stated preference survey sample and Canada as a whole, repeating the survey in multiple centres would nevertheless be desirable, as would collecting illness episode, cost of treatment, and stated preference information from the same sample of respondents.

Response rate is a potential concern for both the cost of illness study and the stated preference survey. Response rates to the emergency department follow-up survey were greater than 80% for individuals with asthma and respiratory infections, but only 50% and 67% for those with COPD and cardiac conditions respectively. Those with cardiac conditions who completed follow-up interviews were younger (and possibly healthier) than those who did not [29]. Cost of treatment estimates may therefore be

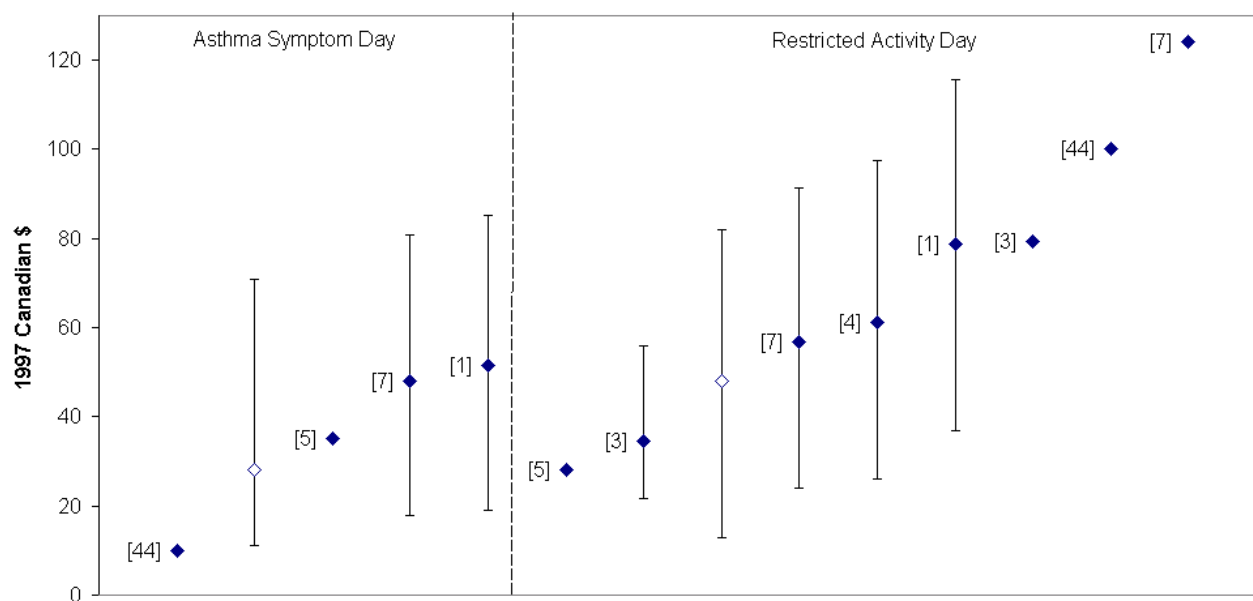


Figure 2

Comparison of valuation estimates for restricted activity days and asthma symptom days with earlier studies. Legend: Labels denote reference number. Open symbols denote estimates from the present study. See Table 6 (additional file 1) for additional details on individual estimates.

conservative relative to a fully representative population of patients of all degrees of severity. Lower and upper bounds on the response rate for the stated preference survey were 22 and 63 percent, assuming that respectively all or none of the telephone numbers which could not be contacted or verified as households were included in the denominator [30]. Of prospective subjects who completed the screening telephone survey, 45 percent agreed to take the in-person survey, while 72 percent of those who scheduled an appointment actually completed the in-person survey [30]. Thus, a substantial portion of the sample did not complete the in-person survey. While a low response rate would raise concerns about the representativeness of the sample, no major differences were observed with respect to demographic characteristics or health status characteristics which were found to influence WTP in this study.

An additional potential limitation of our results is the possibility of double counting of out of pocket expenditures, which could potentially have been captured by both the cost of treatment study and the stated preference will-

ingness to pay study. Ungar et al. [47] found that out of pocket expenses accounted for only 1 per cent of annual per patient treatment costs for asthma, which suggests that this is probably not an important issue.

An illustrative example

We illustrate the application of our approach by examining the benefits associated with reduced cardiorespiratory emergency department visits and hospital admissions attributable to the decline in particulate sulfate concentrations observed in Toronto from the mid 1980s to the late 1990s. Average particulate sulfate concentrations in the Toronto Census Metropolitan Area (CMA) declined from 5.0 $\mu\text{g}/\text{m}^3$ (1984–1986 average) to 3.8 $\mu\text{g}/\text{m}^3$ (1997–1999 average) during this period. We estimated the benefits in 2000 when the estimated population was 4.771 million, using results from an epidemiological study examining the association of air pollution with hospital admissions for respiratory and cardiac disease in southern Ontario [22]. We used this study rather than more recent work because of recently identified problems with the statistical analysis software which has become the standard

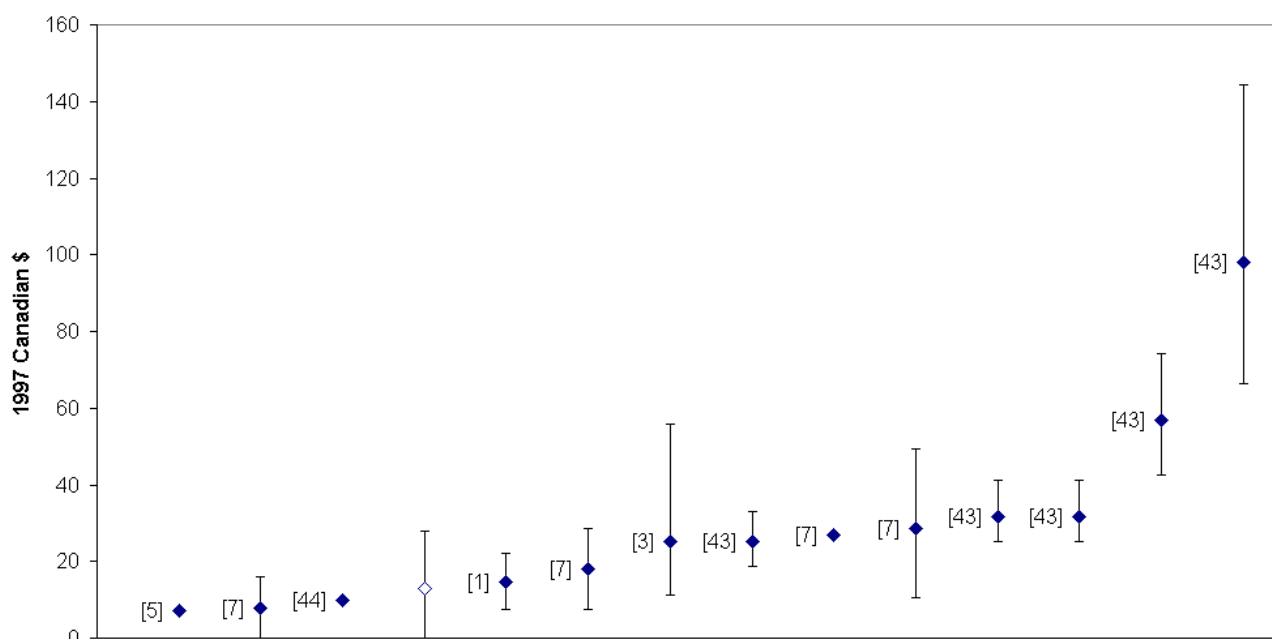


Figure 3

Comparison of valuation estimates for acute respiratory symptom days with earlier studies. Legend: Labels denote reference number. Open symbols denote estimates from the present study. See Table 6 (additional file 1) for additional details on individual estimates.

over the past 5 to 6 years in the analysis of time-series data [52], the full impact of which is still being evaluated. The rates of admission reported in the study translate into 25,000 cardiac and 28,000 respiratory hospital admissions in the Toronto CMA in 2000. Applying the proportion of cardiac and respiratory emergency department visits resulting in admission to hospital from the Saint John study (76% and 20%), [29,33] the numbers of hospital admissions correspond to 33,000 cardiac and 140,000 respiratory emergency department visits in 2000. Based on this study, we estimated that a $1.2 \mu\text{g}/\text{m}^3$ decrease in sulfate concentration would respectively result in a 0.30 % decrease in cardiac hospital admissions/emergency department visits (95% CI 0.16–0.45) and a 0.32% decrease in respiratory hospital admissions/emergency department visits (95% CI 0.21–0.43), accounting for the simultaneous effects of ozone and temperature. The proportional reduction is the same for admissions and emergency visits because in this example the concentration response function is simply scaled based on the relative frequency of the two types of outcome.

We estimated the benefits of the observed changes in sulfate concentration, specifying both the concentration response relationship and the valuation estimates generated above as normal distributions based on the point estimate and standard error, using a Monte Carlo simulation with 1,000 iterations implemented using Analytica software [53]. It was estimated that the benefits of reduced emergency department visits for respiratory and cardiac disease were respectively \$910,000 and \$440,000, while the benefits of reduced respiratory and cardiac hospital admissions were \$380,000 and \$390,000. However, because essentially all non-elective hospital admissions occur through the emergency department, and the valuation estimates generated above for emergency department visits reflect the probability and valuation of subsequent admission to hospital, the total benefits are simply the sum of those related to emergency department visits or \$1.4 million (95% CI 0.91–1.8 million). We emphasize that this estimate does not reflect the much larger benefits attributable to reduced mortality and chronic morbidity.

Conclusions

We believe our overall approach addresses a number of the limitations of the current literature and is widely applicable to future assessments of the benefits of improved air quality. However, because we have noted potentially significant differences between the U.S. and Canada in valuation of certain outcomes, the revised morbidity valuation estimates presented here are most suitable for use in assessing the benefits of air pollution mitigation initiatives in Canada. We recommend employing sensitivity analyses with respect to severity of activity limitation, for those health outcomes where we observed significant sensitivity to this factor.

Abbreviations

COI – cost of illness

COPD – chronic obstructive pulmonary disease

COT – cost of treatment

LP – lost productivity

WTP – willingness to pay

WTA – willingness to accept compensation

Competing interests

None declared.

Authors' Contributions

All authors contributed to drafting of the manuscript and approved the final version. DMS conceived of the study design and contributed to the statistical analysis. PD conceived of the study design. FRJ participated in study design and supervised the statistical analysis. MPM programmed the statistical analysis. AHA participated in study design and the statistical analysis. RCB participated in study design. SJ contributed to the statistical analysis.

Additional material

Additional File 1

See additional file 1, an Adobe Acrobat file, which contains a tabular summary of the properties of valuation estimates utilized in recent benefits assessment studies, as shown graphically in figures 1 to 3.

Click here for file

[<http://www.biomedcentral.com/content/supplementary/1476-069X-1-7-s1.pdf>]

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